

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			2	1			1		
	(b)	(i)		Slows down the neutrons (1) which enables the fission reaction to happen (1)	2			2		
		(ii)		Fission reaction [in fast breeder] happens with fast neutrons / no need to slow neutrons down for fission reaction [so moderator not needed]			1	1		
	(c)			Indicative content: Advantages 1. Underground storage ensures a large distance between HLW and people. 2. Ionising radiation absorbed by the Earth so reduces exposure to people living on the surface. 3. Glass / containers / concrete are stable for thousands of years which reduces the risk of HLW leaking. 4. Employment for scientists/security workers etc who work at the HLW storage facility. 5. Easy to monitor the HLW during long term storage. Disadvantages 1. Need stable geological area to ensure HLW storage is safe. 2. Costly to construct, monitor and ensure high security of deep underground HLW storage area. 3. HLW has long half-life so storage facility needs to operate for 1000's of years. 4. Safety and security risks associated with transporting HLW to the storage facility. 5. NIMBY concerns.	6			6		